

Contact Center Knowledge Assistant — Retail MCP Design

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Design Document — Retail/E-commerce Contact Center Knowledge Assistant

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Design Document — Retail/E-commerce Contact Center Knowledge Assistant

Author: Rishi — Intern 3 (Retail / e-commerce) **Version:** 1.0 (draft for approval) **Due:** end of Wednesday 12 August 2026 (Design-approved gate) **Repos:** code + docs in my own repo `rishiupadhyaygithub/retail-mcp-chatbot`; the shared team repo holds the **contract only** (per manager: code and docs are per-intern, not namespaced in one shared repo)

Legend. Text marked **[ASSUMPTION]** is a guess I will verify — each says how. Text marked **[AGREE]** must be settled jointly with the other three interns (contract v1). Text marked **[TODO]** is a value I fill once I have it (e.g. my machine's IP). Per the brief: a document that labels its uncertainty is stronger than one that performs confidence.

0. Context

An agent is on a live call and needs a grounded answer fast. Some answers come from **documents** (“how long does a refund take?”), some from **records** (“was this order shipped?”), some end in an **action** (“open a return”). I build (1) a Retail MCP server exposing document search, structured queries, and one write action, and (2) a chatbot host that connects to **all four** interns’ servers, routes each question to the right server and the right tool type, and produces a grounded, cited answer — or refuses.

The one rule: my MCP server never calls an LLM. Retrieval and lookup in the server; all reasoning and generation in the host.

Industry assignment (from the brief):

Intern	Industry	Action tool
1	Banking	Raise a transaction dispute
2	Hospitality	Log a guest complaint
3 (me)	Retail / e-commerce	Open a return / RMA
4	Telecommunications	Raise a fault ticket

My domain: documents = order tracking, returns, delivery failures, payments, warranties. records = orders, line items, shipments, returns. action = open a return/RMA.

1. Architecture

COMPONENTS AND WHAT RUNS WHERE

The only shared resource is the LLM. **Ollama runs on the central GB10 server** (`10.10.150.150:11434` , both embeddings and chat). Everything else runs on **my own machine** (`10.10.180.132`), including the retail MCP server (HTTP transport on port **8003**).

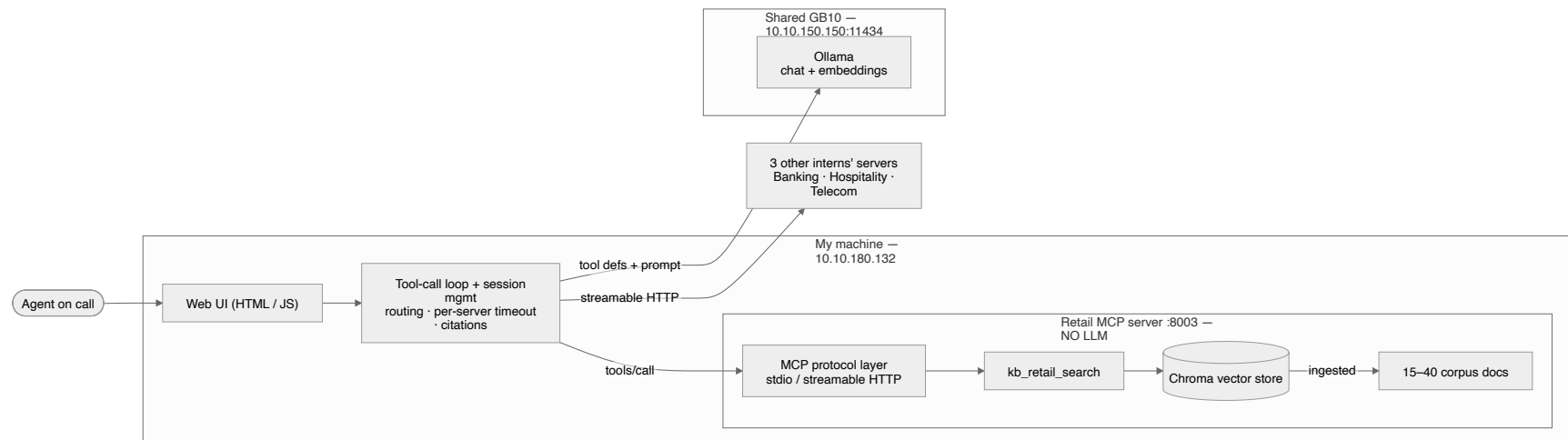


Figure 1 — Phase 1 architecture and data flow (retail)

Figure 1. Solid components exist in phase 1. The server never calls the LLM — retrieval in the server, reasoning in the client.

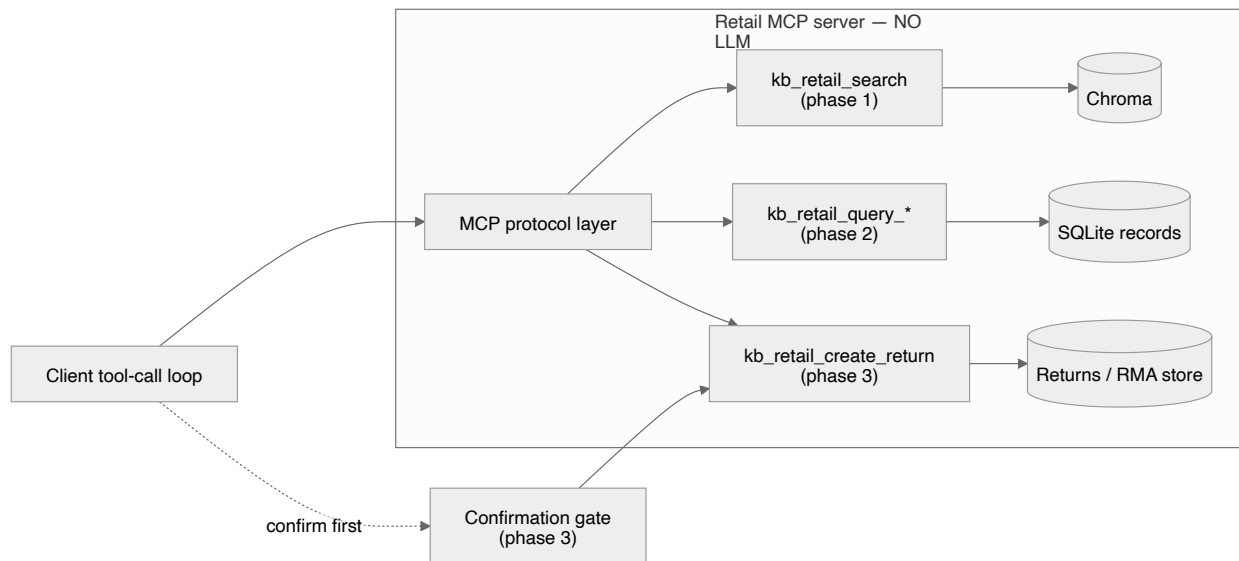


Figure 2 — Where phase 2 (records) and phase 3 (action) attach

Figure 2. Phase 2 (query tools + SQLite) and phase 3 (write action + confirmation gate) attach to the same protocol layer and the same client loop — nothing gets rebuilt.

My host connects to **four** MCP servers (four client sessions): my own + the three others over streamable HTTP.

DATA FLOW FOR ONE QUERY, END TO END

Example: "How long does a refund take, and did order 10231's refund actually go through?" (a composite question).

1. UI sends the user turn to the host.
2. Host builds a prompt: system prompt + conversation so far + the tool catalogue **discovered at runtime** from all four servers.
3. Model emits tool calls. Routing picks the **retail** server, and both tool *types*: `kb_retail_search` (the policy) and `kb_retail_query_orders` (the record).
4. Host runs the calls (in parallel, per-server timeout), gets back passages + rows.
5. Host formats results compactly into the prompt (columnar for rows; strips internal IDs/scores into a citation map).

6. Model produces a grounded answer citing the document and the row. Host renders it with provenance in the UI.

SERVER / CLIENT BOUNDARY (EXPLICIT)

- **Server:** takes a query, retrieves/looks up, returns data + metadata. Never generates prose. Never calls an LLM.
- **Client:** all reasoning — routing, tool selection, multi-round loop, grounding, citation, refusal, confirmation for writes.

WHERE PHASE 2 & 3 TOOLS ATTACH

Phase 1 ships `kb_retail_search` (+ resources + a prompt). Phase 2 adds parameterised **query** tools (`kb_retail_query_orders`, etc.) against a SQLite dataset. Phase 3 adds one **write** tool (`kb_retail_create_return`). All three are the *same server*, discovered by the same client with **no client code change** — that is the point of runtime discovery.

LANGUAGE & STACK, AND WHY

- **Language: Python.** [ASSUMPTION → verify by building the Inspector smoke test day 1] The official MCP Python SDK is mature, Ollama has a clean Python client, and I already proved the tool-call loop in Python (see §4). Rewriting later would cost more than it saves.
 - **MCP server:** official `mcp` Python SDK, exposing stdio + streamable HTTP from one codebase (transport is separable from protocol).
 - **Vector store:** [ASSUMPTION] ChromaDB (local, persistent, metadata filtering, trivial setup). Alternative considered: FAISS (faster, but no built-in metadata filter → more glue). I will confirm after ingesting a real corpus and measuring retrieval latency against the $\leq 300\text{ms}$ target.
 - **Relational store:** SQLite. At a few thousand rows, simpler is better (brief's guidance).
 - **UI:** minimal — see §6.
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2. Protocol understanding

Specification: Model Context Protocol, spec revision **2026-07-28** — team-aligned, all four interns target this version. I cite the spec itself, not just SDK docs. The four points the brief asks for follow.

- **Initialization handshake.** Client and server exchange `initialize` / `initialized`. Each sends `protocolVersion`, `capabilities`, and implementation info. If versions/capabilities don't line up, the session fails here — this is exactly the “initialization failure” I must handle gracefully rather than crash.
- **Capability negotiation.** Both sides declare what they support. My server **declares**: `tools`, `resources`, `prompts` (and `logging` **[ASSUMPTION]**). My client only uses a capability after the server declares it — and must survive a server that *declares* a capability it then doesn't honour (a required robustness case).
- **The three primitives — who decides to invoke:**

Primitive	Invoked by	Mine
Tools	the model, mid-reasoning	<code>kb_retail_search</code> , later <code>kb_retail_query_*</code> , <code>kb_retail_create_return</code>
Resources	the host/user picks	document list; data schema
Prompts	the user, deliberately	one query template

Building only tools would leave me thinking MCP is “REST with a JSON schema attached.” Resources + prompts are a couple of hours each and make the distinction concrete.

- **Runtime discovery.** Client calls `tools/list` (and `resources/list`, `prompts/list`) at session start — nothing hardcoded. A tool renamed or added in v2 is picked up with no client change.
- **MCP Inspector** validates my server before I write any client code. If it fails in Inspector it fails everywhere.

3. Tool-type routing (the hard problem)

By phase 2 the model sees, across four servers, **two tool types each** (search vs query) plus one action — and must pick the right server *and* the right tool type. “How long does a refund take?” plausibly matches all four servers. This is the most common failure the eval set is built to catch.

My approach (considered, not final):

1. **Descriptions do the routing.** The tool `description` is how the model picks both server and tool type, so I write them to disambiguate: `kb_retail_search` → *“Search Retail/e-commerce policy & help documents (returns, delivery, payments, warranties). Use for ‘what is the policy’ questions.”* vs `kb_retail_query_orders` → *“Look up specific Retail orders, shipments and returns by id/customer. Use for ‘what happened on this order’ questions, never for policy.”* Vague descriptions are the main cause of mis-routing.
 2. **System prompt encodes the rule.** “Policy/how-does-it-work → search. Specific account/order/number → query. Change state → action, and only after confirming with the user.” (Full text in §5.)
 3. **Server selection by domain match.** Route on which industry the question is about; a comparative question (“do bank and telco refunds differ?”) fans out to two servers.
 4. **Composite handling.** Questions needing both a document and a record trigger two tool types in one turn (multi-round loop, §5).
 5. **Measured, not asserted.** Routing accuracy (server ≥90%, tool-type ≥90%, spurious ≤1/query) is in the scorecard (§7). **[ASSUMPTION]** description-driven routing hits target on a 7B model; if not, my fallback is a lightweight pre-classification step in the host (still no LLM in the *server*).
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4. Chat model — with evidence

Model: `qwen2.5:7b-instruct` via Ollama on GB10. Chosen because Ollama’s tool-calling varies sharply by model and some advertise it while doing it badly — discovering that in week 3 is fatal — so I verified a full round-trip now. Weaker models tested (mistral, gemma) were unreliable.

Verified tool-call round-trip (retail record lookup):

```

MODEL: gwen2.5:7b-instruct
USER:  How many Wireless Headphones (SKU12345) are in stock, and what's
the price?
TOOL CALL:  lookup_product({'sku': 'SKU12345'})
TOOL RESULT: {'name': 'Wireless Headphones', 'price': 79.99, 'stock': 42}
FINAL ANSWER: There are 42 units of Wireless Headphones (SKU12345) in
stock, at $79.99 each.
PASS: full tool-call loop worked.

```

Script: `client/toolcall_test.py`. **[ASSUMPTION]** the same reliability holds once four servers' tools are in the catalogue at once; I re-test after phase-1 integration. **[AGREE]** the four of us confirm GB10 has this model pulled and that it handles tool-calling — an escalation item if not.

5. Client design (sketch)

THE TOOL-CALL LOOP

```

receive user turn
build prompt = system_prompt + history + runtime tool catalogue (all 4
servers)
loop (max 5 iterations):           # cap so a confused model can't loop
forever
    response = model.chat(prompt, tools)
    if response has tool_calls:
        run calls in PARALLEL, each with a per-server timeout
        for a write tool: PAUSE — confirm with user, show exact fields,
then execute
        append results (compact form) to prompt
        continue
    else:
        return response.content      # final grounded answer

```

WHEN A SERVER IS SLOW OR DOWN

Per-server timeout; on timeout or connection error the host **degrades gracefully** — it returns a partial answer plus a clear note (“Telecom server unreachable; answering from the other three”), never hangs, never crashes. Covers the required cases: one server

down, all down, a server restarting mid-session, malformed tool arguments, a server declaring a capability it doesn't honour.

FIRST DRAFT OF THE SYSTEM PROMPT (ACTUAL TEXT — WILL IMPROVE)

You are a contact-center knowledge assistant. An agent is on a live call.

GROUNDING

- Answer **ONLY** from tool results (retrieved passages or returned rows).
- If the tools return nothing relevant, say you don't know. Never guess a policy, a number, a date, or a status.

TOOL SELECTION

- Policy / how-does-it-work questions -> a **search** tool.
 - Specific order/account/number lookups -> a **query** tool.
 - Requests to change state (open a return, raise a case) -> an **action** tool,
and **ONLY** after confirming the exact fields with the user.
 - Pick the tool whose description matches the industry in the question.
- For a
comparison across industries, call more than one server.

CITATIONS

- Every claim names its source: the server and the document or the record (table + row) it came from.

REFUSAL

- If asked something the tools can't answer, refuse plainly. A refusal is a
correct answer; a confident wrong answer is the worst outcome.

WRITES

- Never call a write tool unattended. State the fields you will submit and ask
the user to confirm before executing.

Context budgets and prompt formatting are deferred to the addendum (once I've measured real chunks), per the brief.

6. UI

A single **plain HTML/JS page** served by the host — no framework, no build step (chosen to match the team and keep setup near-zero; a static page that `fetch` es the host is faster to stand up than any toolkit and gives full control over how provenance renders). It replaces the console, nothing more. One live multi-turn conversation; **no persistence** (reload = fresh); shows **provenance** (which server + which document/record per claim) beneath each answer; grows one step per phase (records display at phase 2, confirmation dialog at phase 3). No styling system, no component libraries. Weighed on setup cost, as instructed — this must work from the end of phase 1 and is not where time should go. Alternative considered: Streamlit (fast, but Python-server-bound and off the team's stack) — rejected for team consistency.

7. Evaluation plan

Eval set + harness exist before the chatbot does (baseline gate). 25–30 hand-written questions, phrased the way an agent asks mid-call, each recording what *should* happen (which docs / which query / which action). Categories: document (incl. two-document answers and vocabulary differing from source), record (incl. an aggregate), composite, cross-server (≥ 2 needing another industry + one comparative across two), action (incl. one missing-field $\rightarrow$ client must ask), unanswerable. Lives in `eval/`.

Scorecard reported twice — a baseline as soon as retrieval works, and a final set after tuning. Each layer measured separately (a wrong final answer could be retrieval, routing, query, or the model ignoring good data):

Layer	Metrics & targets
Retrieval	Recall@5 $\geq 85\%$, Recall@1 $\geq 60\%$
Routing	server $\geq 90\%$, tool-type $\geq 90\%$, spurious $\leq 1/\text{query}$, cross-server synth $\geq 80\%$, composite $\geq 80\%$
Structured	query correctness $\geq 90\%$, numerical accuracy 100%, empty-result honesty 100%
Answer quality	groundedness 100%, citation accuracy 100%, correct refusal 100%, false refusal $\leq 10\%$
Action safety	spurious writes 0, fabricated fields 0, action routing 100%, confirmation shown 100%
Latency	e2e p50 $\leq 4\text{s}$, p95 $\leq 10\text{s}$, retrieval $\leq 300\text{ms}$, query $\leq 100\text{ms}$, MCP overhead $\leq 100\text{ms}$ (warm vs cold reported separately — Ollama unloads idle models)
Token	tokens/query reported then reduced $\geq 40\%$ vs naive raw-JSON injection, no quality regression
Robustness	pass/fail: 1 server down, all down, empty query, 5000-word query, wrong-language query, off-topic, 10000-row match, missing customer ref, two concurrent identical requests — none may crash

Harness: one script (`eval/run_evals.py`), one command, prints the scorecard as one table; reads the eval set, runs each question in a **fresh session**, records per-layer numbers, instruments token counts from the baseline run onward.

Targets I may argue against [ASSUMPTION]: Recall@1 $\geq 60\%$ may be tight if my corpus has genuinely conflicting policies (the brief *wants* contradictions) — a passage can be “correct” two ways. If baseline shows this, I’ll propose Recall@1 measured against an accept-set of valid passages, with reasoning, rather than silently missing target.

8. Risks

Risk	Likelihood	Plan
Tool-type routing below target (hardest problem)	High	Description-driven first; fallback host-side pre-classifier; measured early
GB10 / Ollama unreachable or model can't tool-call	Med	Escalate immediately (brief §13); verified round-trip already done
Four-way contract disagreement	Med	Bring a concrete draft (contract v1) to shorten debate; escalate if no convergence
Retrieval quality on messy/contradictory corpus	Med	It's a bottomless pit — cap tuning time; report honestly
My machine off during interop/demo → 3 others blocked	Low/High-impact	Keep awake + on network for all shared dates; verify reachability before interop
Cold-start latency mistaken for bugs	Med	Report warm vs cold separately; stagger bulk ingestion (Ollama shared)

Least-confident estimate: **the client's tool-call loop + four-server interop** (§9). If it slips, phase 1 demo slips.

9. Phase 1 in detail

Corpus (sources). 15–40 real public documents from **four** retailers chosen for *genuinely conflicting* policies — **Amazon, Best Buy, IKEA, Target** — drawn from their help centres and returns / warranty / delivery / payments / order-tracking pages. Public pages only; every source URL is listed in `data/sources.md`. **[ASSUMPTION: exact doc list pinned at ingestion.]**

- **Deliberate contradiction pair:** Amazon's ~30-day standard return window vs Best Buy's 15-day window with an electronics restocking fee — the system must surface the conflict, not blend the two into one wrong number.

- **Vocabulary deliberately differs** across companies: “return” vs “RMA” vs “exchange”; “refund” vs “money back”; “carrier” vs “courier”. Retrieval must match on meaning, not string overlap.
- **Mixed length:** short (a single account-return FAQ, ~60 words) through long (a full terms-of-sale / limited-warranty disclosure, several thousand words across many subsections) — uniform length hides chunking bugs.
- **Language:** all sources are English (US retailers). None non-English is required; I run one non-English query at baseline to confirm bge-m3 degrades gracefully rather than crashing.

9. Ingestion & chunking. Split by section (markdown `##` heading), not fixed token count. Every sourced doc is saved as markdown with `##` marking natural topic boundaries (an individual FAQ, a named warranty clause). Splitting on those boundaries keeps each chunk topically self-contained, which is what makes a citation trustworthy — the `section` field maps 1:1 to a real heading. **Overlap:** none between sections (boundaries are semantic, not arbitrary); for any section over ~400 words (the long disclosure docs), a secondary split with ~50-word overlap so no clause is orphaned.

Metadata per chunk: `source` (doc title), `section` (heading text), `chunk_id`, `document_type` ∈ {`returns`, `warranty`, `delivery`, `payments`, `order_tracking`}.

Why not fixed-token: a fixed size would cut the short FAQ needlessly and slice mid-clause through the long disclosures — the brief flags uniform chunking as the weak default. **Verify:** tune on real Recall@5; if heading-split underperforms on the long docs, fall back to heading-then-token hybrid before phase 2 adds token pressure.

10. Vector store & data model. ChromaDB (local, persistent on disk). One collection (`retail_docs`); schema per record: {`content`, `embedding`, `source`, `section`, `chunk_id`, `document_type`}. **chunk_id format** `retail-doc-<n>:chunk-<n>` (e.g. `retail-doc-3:chunk-12`) — the exact contract-v1 format so citations are portable across all four servers. Indexing: HNSW (Chroma default; negligible impact at <100 chunks). Similarity: **cosine, scores normalized 0–1** before return (contract v1 — comparable across all four servers regardless of store). Why Chroma: metadata filtering + persistence + near-zero setup; weighed against FAISS (faster, no built-in metadata filter → more glue).

11. Embedding model. `bge-m3` via Ollama on GB10 (the brief's suggested default). Why: strong retrieval quality and multilingual — though my corpus is all English (US retailers), so the multilingual capacity is headroom, not a current requirement. I run one non-English query at baseline to confirm it degrades gracefully. **[ASSUMPTION]** Alternative held in reserve: `nomic-embed-text` (lighter) — switch only if `bge-m3` misses the $\leq 300\text{ms}$ retrieval target or underperforms on measured Recall@5.

12. Retrieval strategy. `top_k` default 5 (matches contract). Metadata filtering available (e.g. by `source`). **No-match handling:** below a similarity floor **[ASSUMPTION: threshold tuned on baseline]**, return an empty `results` array — a success, so the host can pass “found nothing” to the model verbatim and refuse.

13. Failure handling. Server down → host degrades (partial + note). Empty retrieval → refuse, don't invent. Model answering without evidence → system prompt forbids it and groundedness scoring (100% target) catches it.

10. Phases 2 & 3 — outline (half page)

Phase 2 (records). Data I'll need: ~4–6 SQLite tables — `orders`, `line_items`, `shipments`, `returns`, `customers` — a few hundred to a few thousand rows, **consistent with my documents** (if a doc says refunds take 5 working days, the data shows ~5, with a few exceptions). Include awkward cases: a duplicate charge, a partially shipped order, a return already under review, a customer with two accounts. 3–4 documented reference customers for demos. Tools (parameterised, **not** text-to-SQL): e.g. `kb_retail_query_orders(customer_ref, from_date, to_date, status)`. Results capped + paginated with a truncation flag. What would change my phase-1 design if wrong: the citation format must already carry table+row provenance, so I design §1's citation map to hold record refs now.

Phase 3 (actions). One write tool: `kb_retail_create_return(order_id, line_item_id, reason)` → returns a return/RMA reference. Missing required field **fails loudly**, never defaults. Client collects fields, confirms with the user showing exact fields, then executes. What would change phase-1: the host's loop already has the confirm-before-write pause (§5).

Full schemas, result caps, confirmation flow, context budget, and prompt formatting go in the **design addendum** (before phase-2 build, with contract v2).

11. Plan (this is a graded deliverable)

Two dates are fixed; I plan the rest at half-day granularity, with slack labelled as slack.

Fixed: Task starts Fri 7 Aug 2026 · Design doc + contract v1 due **end of Wed 12 Aug 2026**.

Gates in order (sequence fixed; dates chosen — shared ones marked [AGREE]):

Gate	Meaning	Target date
Design approved	This doc approved; no code before it	Wed 12 Aug
Baseline scorecard	Eval set + harness done, first numbers — before the server	[TODO] ~Fri 15 Aug
Interop day (v1)	All 4 servers live, cross-tested vs contract v1	[AGREE — shared]
Phase 1 demo	Working document chatbot in the UI	[AGREE — shared]
Phase 2 & 3 demo	Records, composite, a case raised end to end	[AGREE — shared]
Final demo + retro	Everything + before/after numbers	[AGREE — shared]

Effort per phase, and what drives it (half-day units, [ASSUMPTION] — my least-confident estimates):

- **Eval set + harness — 1.5 days.** Driven by hand-writing 25–30 questions with known answers and a one-command harness. Gate before server.
- **Corpus + ingestion — 1 day.** 15–40 real public retail docs (3–4 companies, deliberately messy, ≥1 contradicting pair), chunk + embed. Ollama shared → stagger my bulk run.

- **Phase 1 server (search + resources + prompt) + Inspector — 1.5 days.** Both transports; must pass Inspector and run in a third-party host unmodified.
- **Client + tool-call loop + 4-server interop + UI — 2.5 days.** Known-hard; where time disappears; capped.
- **Phase 2 dataset + query tools — 2 days.** Dataset ~1 day; query tool schemas need **[AGREE]** across all four (contract v2).
- **Phase 3 write tool + confirmation — 1 day.**
- **Tuning + token reduction + final scorecard + retro — 1.5 days.**

Dependencies (can't parallelise out of these): ingestion → retrieval → MCP server → client → phases 2 & 3. Blocked-on-others: contract v1 (all four), interop day + all demos (shared, machines up), GB10/Ollama availability.

Where my risk is: the client + interop estimate (2.5d) is least confident. If it slips, I flag it the day I know, not on the gate — a silent slip makes three other people miss a gate too.

12. Assumptions register (summary)

Every **[ASSUMPTION]** above, and how I resolve it: stack (build Inspector test day 1), vector store & embedding (measure recall on real corpus), chunk strategy (heading-split; tune / fall to hybrid on Recall@5), routing method (measure early, host-side fallback ready), Recall@1 target (argue accept-set if contradictions bite). Every **[AGREE]** is a contract-v1 / shared-schedule item. Resolved values now baked in: spec 2026-07-28, GB10 Ollama 10.10.150.150:11434, my host 10.10.180.132:8003, UI plain HTML/JS. Only remaining **[TODO]**: the shared dates (interop + three demos), filled once the team fixes them.

End v1.0. Peer-review with the other three before submitting. Expect one revision round — do not submit at the last hour.